

AI/ML Internship – Day 77 Notes & Tasks

Module 9: Metadata Filtering & Source Citation in RAG – Making AI Responses Accurate and Trustworthy

Part 1: Recap of Day 76

Yesterday we learned:

-  Multi-PDF RAG
-  Loading Multiple PDFs
-  Multiple Document Retrieval
-  Handling Duplicate Information
-  Multi-PDF Question Answering

Today we will learn:

- What is Metadata?
 - What is Metadata Filtering?
 - Why Metadata is Important
 - Source Citation
 - Filtering Documents
 - Building Trust in AI Responses
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Part 2: What is Metadata?

Definition

Metadata is "data about data."

It contains additional information about a document but is **not part of the actual content**.

Simple Definition

Think of a library.

Every book has:

- Book Name

- Author
- Publisher
- Category
- Year
- Shelf Number

These details are metadata.

The actual pages of the book are the document content.

Example

Suppose we upload:

Employee_Handbook.pdf

Metadata:

Filename : Employee_Handbook.pdf

Page : 12

Department : HR

Category : Policies

Language : English

Created Date : 15-05-2026

The document text is different from the metadata.

Part 3: Why is Metadata Important?

Imagine a company has **10,000 PDFs**.

A user asks:

"What is the leave policy?"

Without metadata:

The AI searches every document.

With metadata:

The AI can search only:

- HR Department
- Leave Policy Documents

This makes retrieval:

- Faster
- More accurate
- More efficient

Part 4: Types of Metadata

Common metadata fields:

Metadata	Description
Filename	Name of the document
Page Number	Page containing the text
Author	Document creator
Department	HR, Finance, Legal, etc.
Category	Policy, Manual, Guide
Date	Creation or modification date
Language	English, Malayalam, Hindi

Part 5: What is Metadata Filtering?

Definition

Metadata Filtering limits the search to documents that match specific metadata values.

Example

Company has:

- HR.pdf
- Finance.pdf
- Medical.pdf

- Legal.pdf

User asks:

"What is the leave policy?"

Instead of searching all documents:

Retriever searches only:

Department = HR

This is Metadata Filtering.

Part 6: Metadata in ChromaDB

When storing documents, metadata can be added.

Example

```
documents = [  
    Document(  
        page_content="Internship duration is 3 months.",  
        metadata={  
            "source": "Internship.pdf",  
            "page": 5,  
            "department": "Training"  
        }  
    )  
]
```

Each document chunk now stores:

- Source file
 - Page number
 - Department
-

Part 7: Filtering by Metadata

Example:

```
results = vector_db.similarity_search(  
    "Leave policy",  
    filter={  
        "department":"HR"  
    }  
)
```

Only HR documents are searched.

📌 Part 8: What is Source Citation?

Definition

Source Citation tells users **where the answer was found**.

Instead of only giving an answer, the chatbot also displays:

- File name
 - Page number
 - Document title
-

Example

User:

What is the internship duration?

AI Response:

The internship duration is 3 months.

Source:

Internship_Guide.pdf

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Users can verify the answer.

📌 Part 9: Why Source Citation Matters

Without citations:

✗ Users don't know if the answer is correct.

With citations:

- ✓ Easy to verify.
 - ✓ More trustworthy.
 - ✓ Useful for legal and medical applications.
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📌 Part 10: Enterprise Example

A bank uploads:

- Loan Policy.pdf
- RBI Guidelines.pdf
- Employee Rules.pdf

Customer asks:

"What is the minimum age for a home loan?"

AI replies:

Minimum age is 21 years.

Source:

Loan Policy.pdf

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This builds confidence.

📌 Part 11: Benefits of Metadata Filtering

- ✓ Faster retrieval
- ✓ Better accuracy
- ✓ Less irrelevant information
- ✓ Better enterprise search

✓ Reduced LLM token usage

✦ Part 12: Challenges

- ✗ Incorrect metadata leads to wrong search results.
 - ✗ Missing metadata reduces retrieval quality.
 - ✗ Metadata must be updated when documents change.
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✦ Part 13: Real-World Applications

Legal AI

Search only legal contracts.

Hospital AI

Search only patient guidelines.

University Chatbot

Search only academic regulations.

HR Assistant

Search only HR policies.

Banking Assistant

Search only financial documents.

✦ Part 14: Best Practices

- Always store source file names.
- Store page numbers.
- Use meaningful document categories.

- Keep metadata consistent.
 - Display citations with every answer.
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Part 15: Theory Questions

1. What is Metadata?

Metadata is additional information that describes a document, such as its file name, page number, author, or category.

2. What is Metadata Filtering?

Metadata Filtering restricts document retrieval to documents matching specific metadata values.

3. Why is Source Citation important?

It allows users to verify the source of the AI's answer and increases trust.

4. Give three examples of metadata.

- Filename
 - Page Number
 - Department
-

5. Why do enterprise AI systems use metadata?

To improve retrieval speed, accuracy, and document organization.

Part 16: Interview Questions

What is metadata in a RAG system?

Metadata is additional information stored with document chunks that helps identify and filter documents during retrieval.

Why is metadata filtering useful?

It narrows the search to relevant documents, making retrieval faster and more accurate.

Why should AI systems display source citations?

Source citations allow users to verify answers, increase transparency, and improve trust.

Can metadata affect retrieval performance?

Yes. Accurate metadata improves retrieval quality, while incorrect or missing metadata can lead to poor search results.

Name some common metadata fields.

- Filename
 - Page number
 - Author
 - Category
 - Department
 - Date
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Tasks for Day 77

Theory

1. Define Metadata.
 2. Explain Metadata Filtering.
 3. Why is Source Citation important?
 4. List five metadata fields.
 5. Explain the role of metadata in enterprise RAG systems.
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Practical Task 1

Create five sample documents with metadata including:

- Filename

- Page number
 - Department
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Practical Task 2

Store the documents in ChromaDB with metadata.

Practical Task 3

Retrieve only HR documents using metadata filtering.

Practical Task 4

Ask questions and display the source file name and page number with each answer.

Practical Task 5

Compare retrieval results:

- Without metadata filtering
- With metadata filtering

Write your observations.